Classical ML vs Deep Learning

Machine learning on paediatric pneumonia X-rays dataset

HOG + SVM vs ResNet50

University of Essex Online
Machine Learning April 2026
Assignment: Development Individual Project: Presentation

Significance and motivation

700,000+

deaths in children under five each year

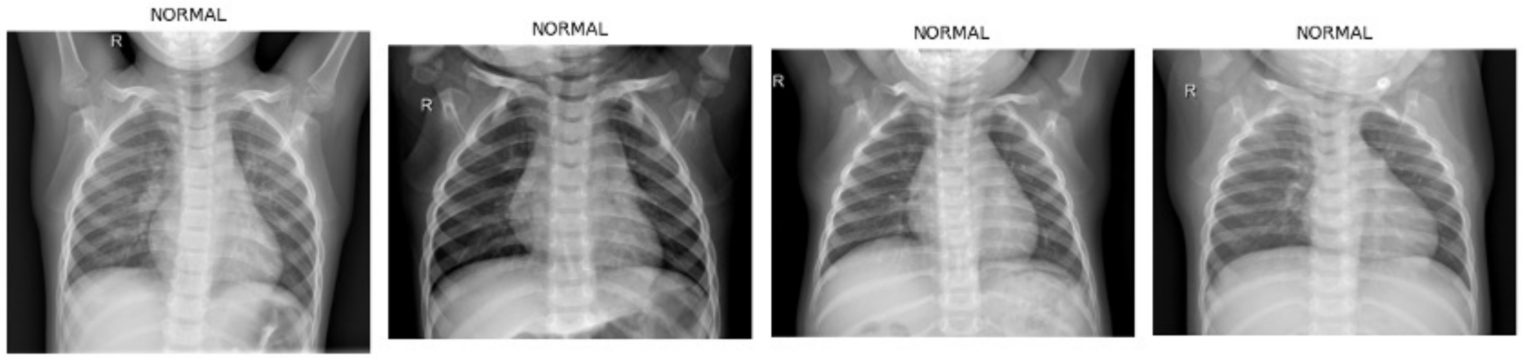
- Specialist assistance, first screening
- Priority: minimise missed pneumonia
- False negative cost > false positive cost
- Risk: models may learn shortcuts

Sources: Topol (2019); UNICEF (2024); Zech et al. (2018).

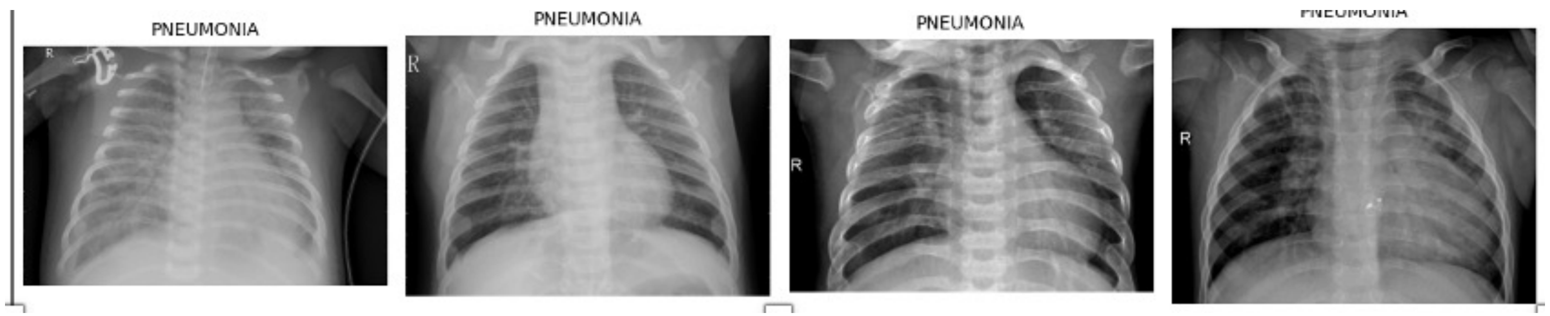
Dataset: paediatric chest X-rays

5,856	3:1	1 hospital
Dataset images	PNEUMONIA : NORMAL ratio	paediatric, single-site - constrains generalisation

NORMAL



PNEUMONIA



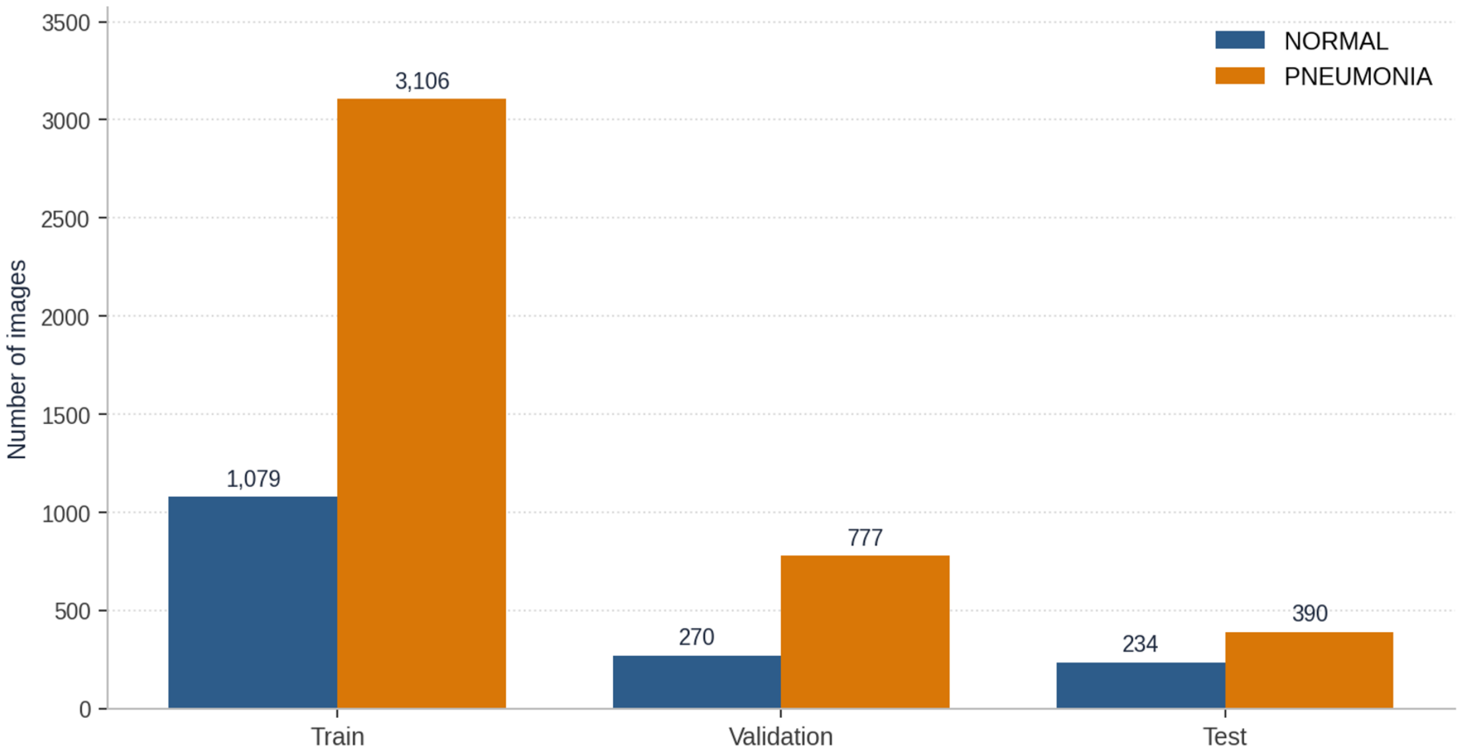
Sample chest X-rays from EDA

- Paediatric NORMAL vs PNEUMONIA classification
- 3:1 imbalance -> macro-F1 and class weighting
- Single hospital -> limited generalisation
- Text and artifacts in the images may create shortcut features

Sources: Kermany et al. (2018); Mooney (2018); Zech et al. (2018).

Class imbalance and split

Class balance



3:1 Imbalance
Balanced class weights in both models

Validation fix
Original validation set: 16 images
New stratified 80/20 split

Robustness check
5-fold stratified cross-validation

Final test
624 images held out
Post-hoc diagnostics

Sources: Fawcett (2006); James et al. (2023); King and Zeng (2001); Kohavi (1995); Prechelt (1998); Saito and Rehmsmeier (2015); Sokolova and Lapalme (2009).

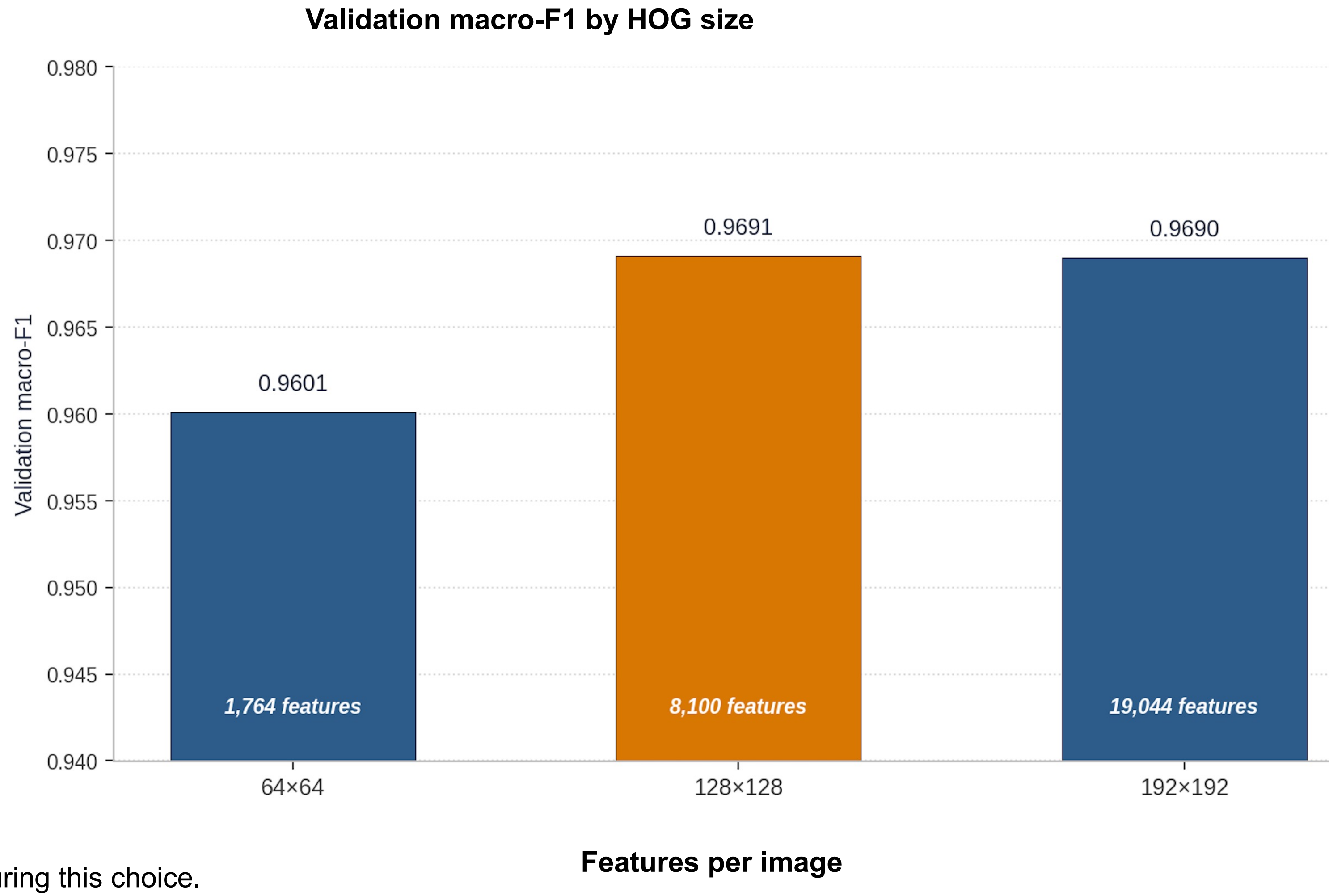
Two preprocessing workflows

Classical: HOG + SVM

X-ray input → Resize 128×128 → Grayscale → Scale /255 → HOG edge features → StandardScaler → SVM (RBF kernel)

Deep learning: ResNet50

X-ray input → Resize 224×224 → RGB (3-channel) → ImageNet preprocess → ResNet50 base (frozen) → Custom head → Sigmoid -> PNEUMONIA score



HOG image size chosen on validation data only. Test set untouched during this choice.

Sources: Dalal and Triggs (2005); He et al. (2016); Shorten and Khoshgoftaar (2019).

Classical model: HOG + SVM

X-ray (128×128, gray, scaled /255) → HOG features → StandardScaler → RBF SVM → Platt scaling → PNEUMONIA score

Features

- HOG captures local edge and shape patterns
- 128×128 grayscale -> 8,100 features
- StandardScaler before classification

Model

- Class-weighted RBF SVM
- C = 1 ; gamma = "scale"
- Platt scaling gives probability-like scores

Chosen as an established non-linear classical baseline.

Sources: Cortes and Vapnik (1995); Dalal and Triggs (2005); Hsu, Chang and Lin (2003); Pedregosa et al. (2011); Platt (1999).

Deep-learning model: ResNet50

Input 224×224×3 → ResNet50 base [Frozen] 23M → Global Avg Pool → Dropout 0.3 → Dense 256 ReLU → Dropout 0.5 → Dense 1 Sigmoid

Greyed = frozen (no gradient updates). White = trainable head.

Training

- Frozen ImageNet ResNet50 base
- Adam + class-weighted loss
- Early stopping on validation AUC

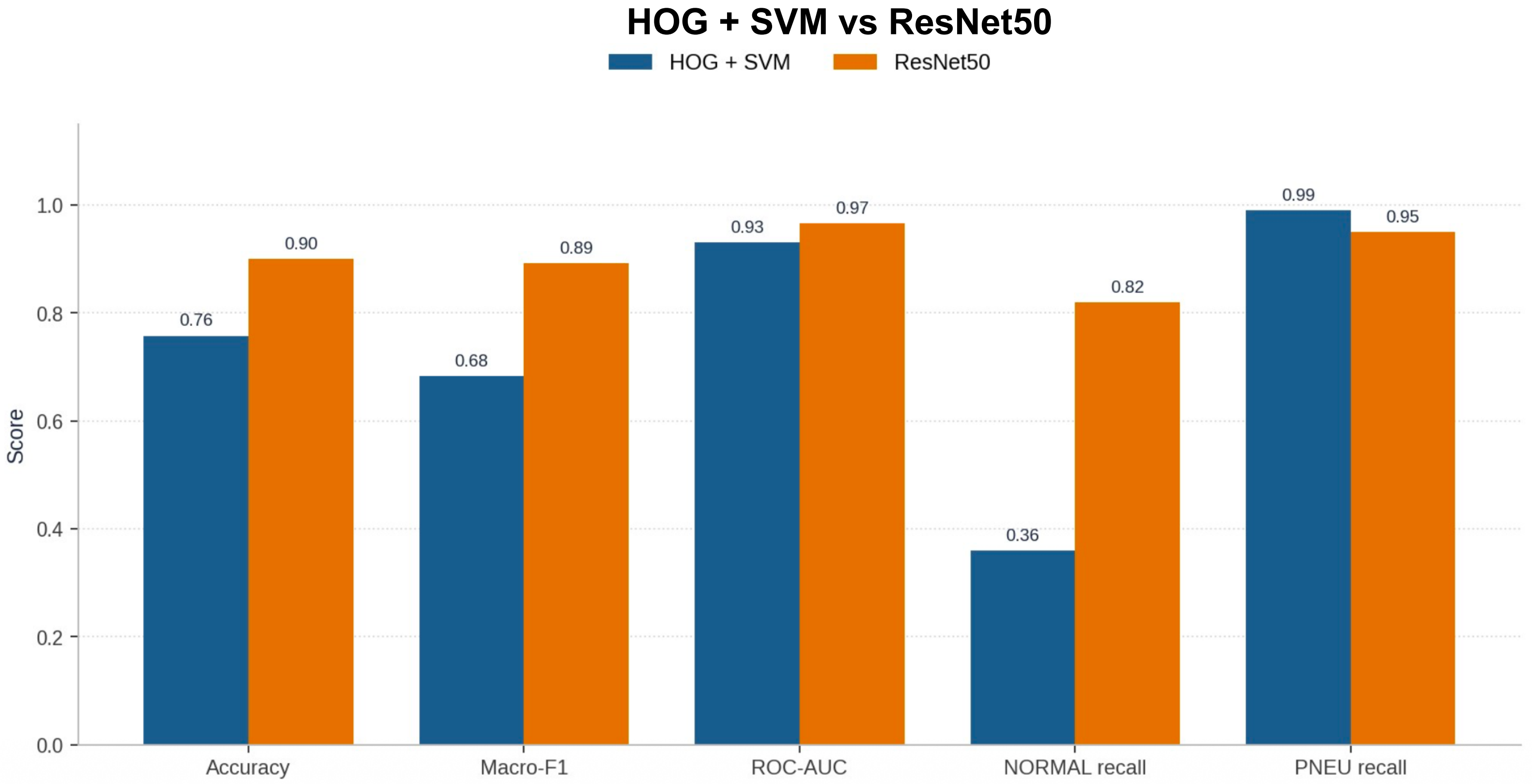
Design choices

- Transfer learning for limited data and compute
- DenseNet121: next comparison
- Fine-tuning attempt did not unfreeze the base

Sources: Cheplygina, de Bruijne and Pluim (2019); He et al. (2016); Kingma and Ba (2015); Prechelt (1998); Srivastava et al. (2014); TensorFlow (2024); Yosinski et al. (2014).

Test-set comparison (held-out 624 images)

Metric	HOG + SVM	ResNet50	Gap (pts)
Macro-F1	0.683	0.892	+21
Accuracy	0.758	0.901	+14
Macro precision	0.850	0.901	+5
Macro recall	0.679	0.885	+21
ROC-AUC	0.930	0.967	+4
NORMAL recall	0.36	0.82	+46



Headline test result

Macro-F1: 0.892 vs 0.683

Generalisation check

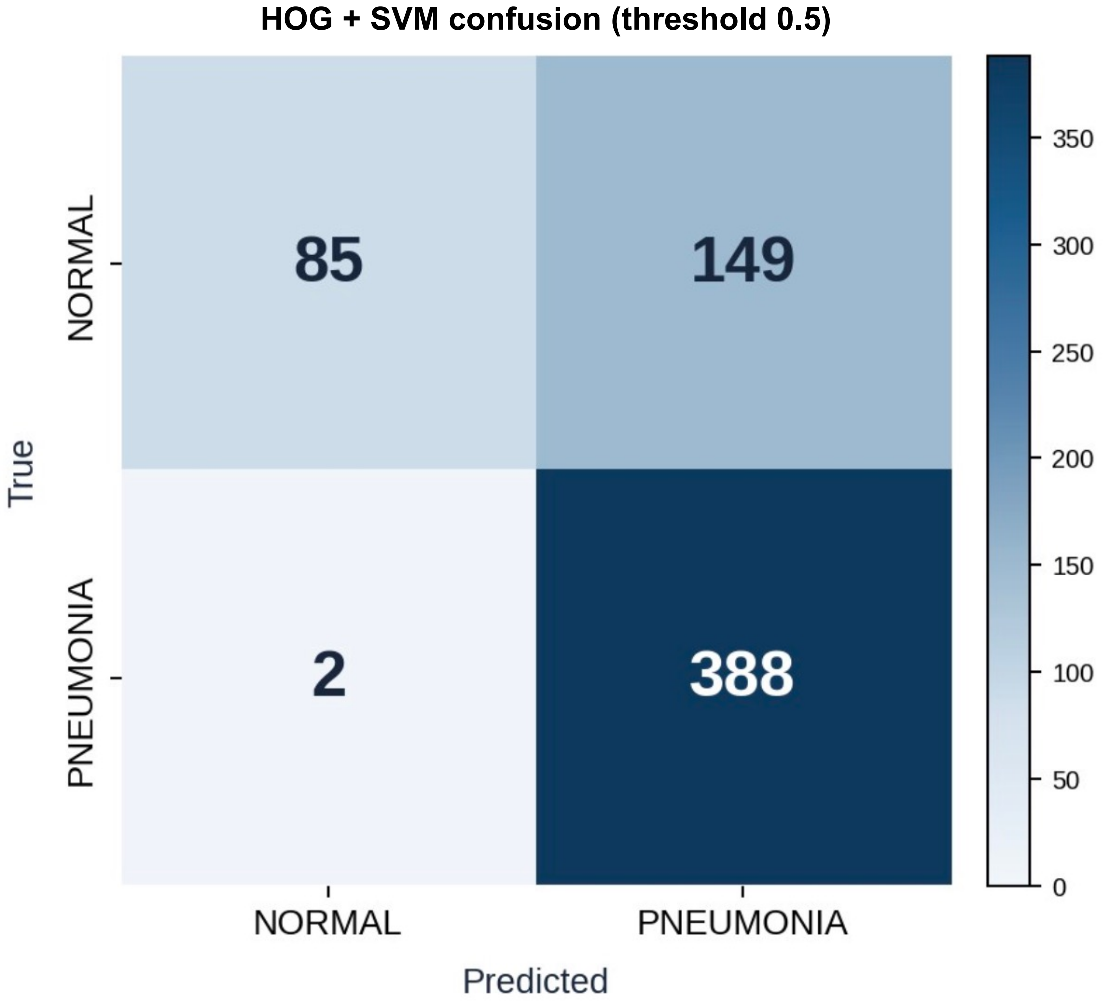
CV -> test macro-F1:
SVM 0.966 -> 0.683
ResNet50 0.902 -> 0.892*

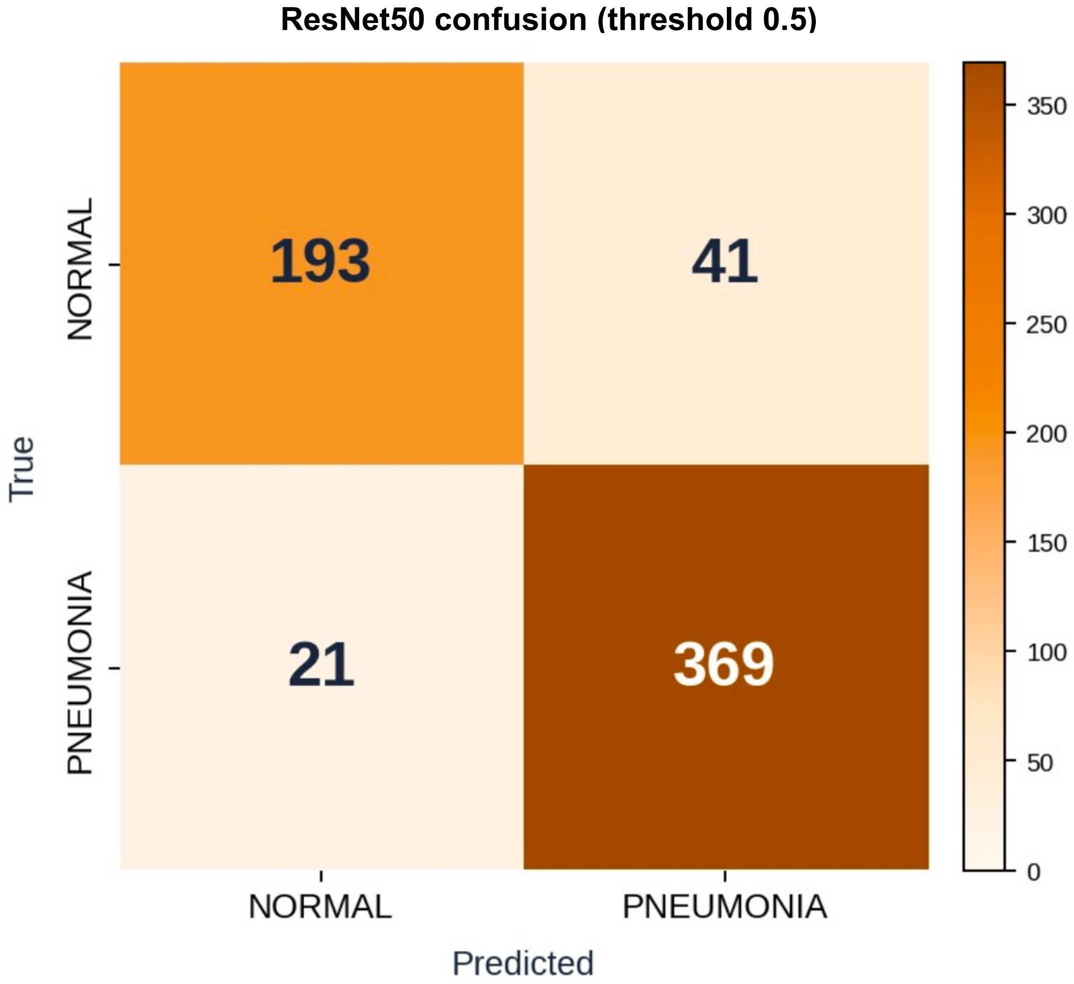
Class-level contrast

NORMAL recall:
0.82 vs 0.36

Sources: Fawcett (2006); Saito and Rehmsmeier (2015); Sokolova and Lapalme (2009).

Confusion matrices





Threshold 0.5

SVM: 149 false alarms; 2 missed pneumonia cases

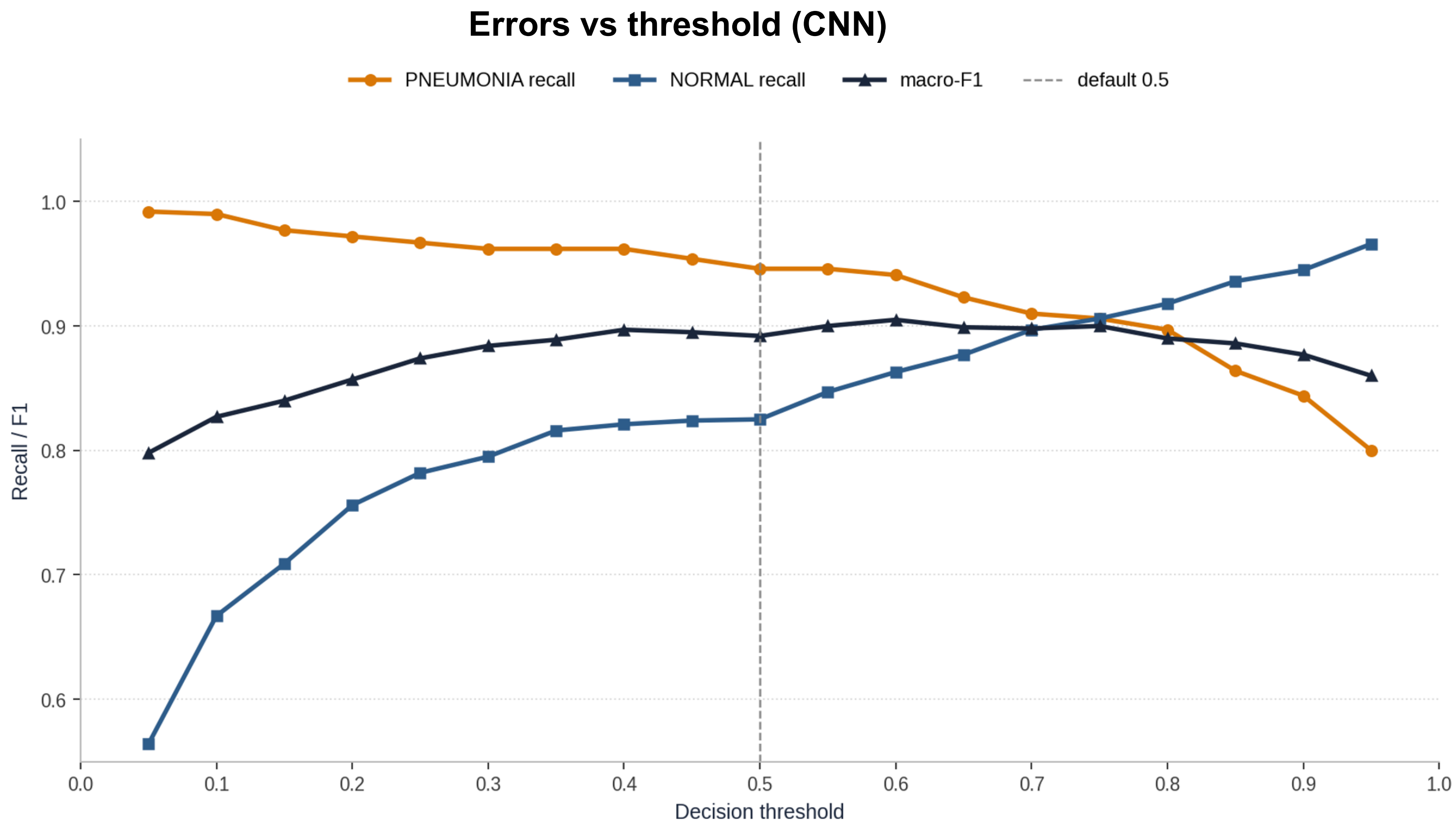
ResNet50: 41 false alarms ; 21 missed pneumonia cases

0.5 is not automatically the clinical threshold.

Sources: Provost and Fawcett (2001).

Exploratory threshold trade-off

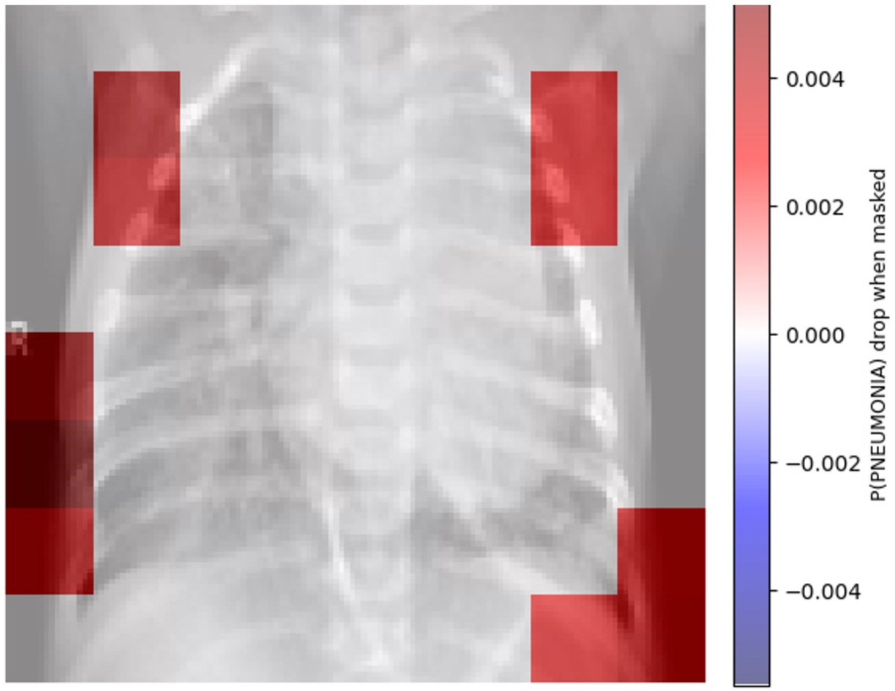
Test-set sensitivity analysis - no deployment cut-off selected



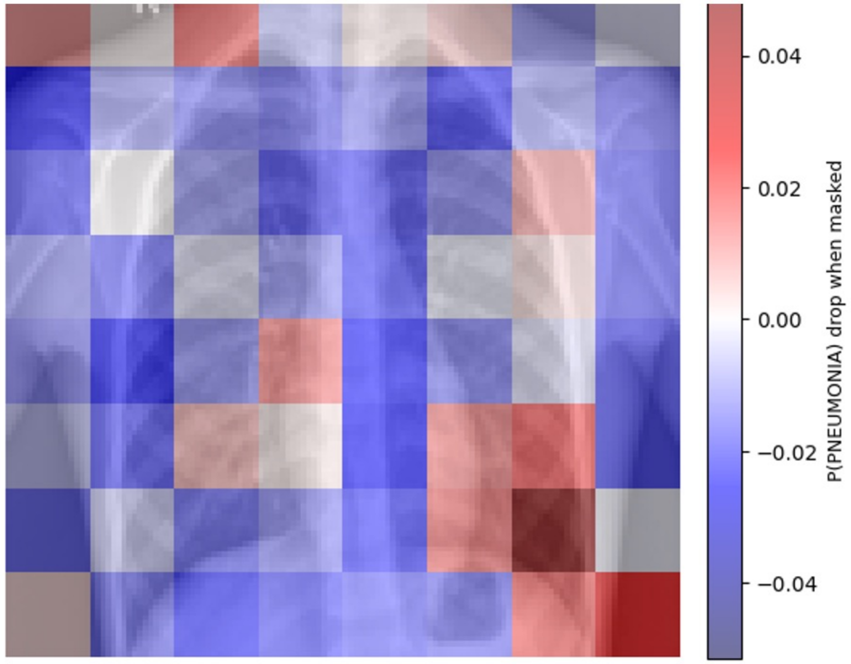
- 0.05 -> PNEU recall 0.992 ; 102 FP ; 3 FN
- 0.50 -> PNEU recall 0.946 ; 41 FP ; 21 FN
- 0.80 -> PNEU recall 0.897 ; 19 FP ; 40 FN
- Lower cut-off = fewer misses, more false alarms
- Final cut-off: select on separate clinical validation data

SVM explanation: patch test on a correct vs misclassified case

Correct pneumonia: patch-test heatmap



Misclassified NORMAL: patch-test heatmap



Mask one patch and measure change in pneumonia score

Correct PNEUMONIA: near-saturated score -> little visible change.

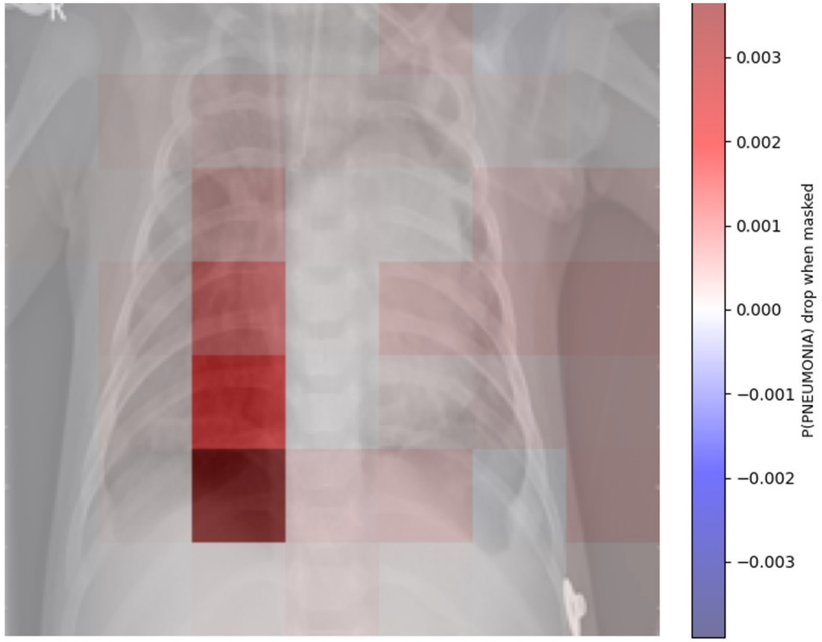
Misclassified NORMAL: score sensitivity near shoulders and image edges.

Shortcut warning, not causal proof. High scores indicate confidence but are not calibrated as probabilities, this is a post-hoc finding and noted as improvement next.

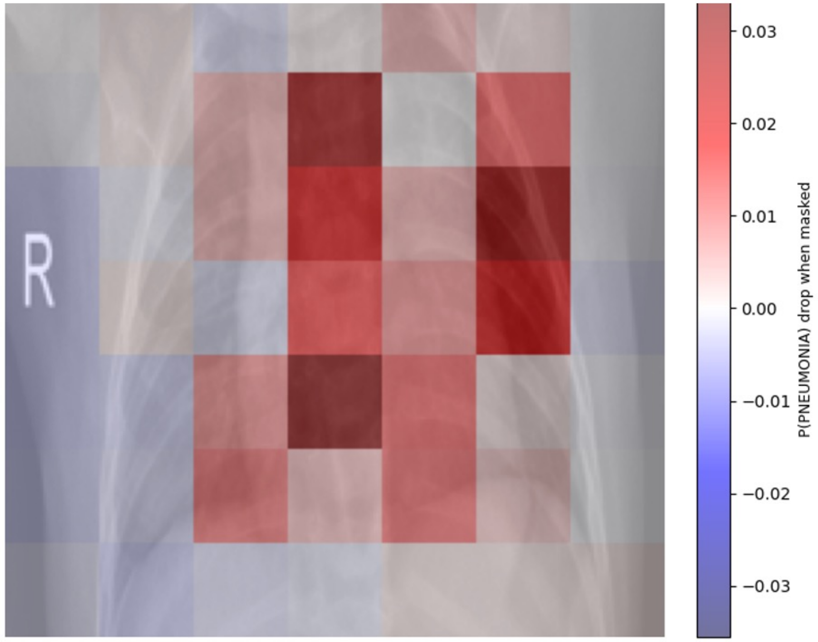
Sources: Guo et al. (2017); Zech et al. (2018); Zeiler and Fergus (2014).

CNN explanation: patch test

Correct pneumonia: occlusion heatmap



Misclassified NORMAL: occlusion heatmap



Same occlusion test as SVM

Scores above 99% (requires calibration)

Misclassified NORMAL: score sensitivity near shoulders and edges

Test Grad-CAM for faster routine display

Sources: Adebayo et al. (2018); Guo et al. (2017); Selvaraju et al. (2017); Zeiler and Fergus (2014).

Demonstration on a held-out NORMAL X-ray

14. Demonstration

```python
# Demo 1 pre-selected image. rational and design in deck.

from pathlib import Path
import numpy as np
from PIL import Image
from skimage import transform
from skimage.feature import hog
from tensorflow.keras.applications.resnet50 import preprocess_input

# Load both saved models from Drive (fast - they are already trained).
svm    = joblib.load(MODEL_ASSETS_OUTPUT / "svm_rbf_v3.pkl")
scaler = joblib.load(MODEL_ASSETS_OUTPUT / "scaler_v3.pkl")
cnn    = load_model(MODEL_ASSETS_OUTPUT / "cnn_resnet50_v3.keras")

# The held-out NORMAL X-ray we will run both models on.
DEMO_IMAGE = DRIVE_BASE / "demo_image.jpeg"
print(f"Demo image: {DEMO_IMAGE.name}")
print(f"True label: NORMAL")
print()

# Run the complete SVM on the image for demo (preprocess, HOG, scale, predict).
# Preprocessing matches the training helper: PIL grayscale, resize to 128, divide by 255.
img_gray = Image.open(DEMO_IMAGE).convert("L").resize((128, 128))
img_gray = np.array(img_gray) / 255.0
hog_feat = hog(img_gray, orientations=9, pixels_per_cell=(8, 8),
               cells_per_block=(2, 2), block_norm="L2-Hys", feature_vector=True)
svm_prob = svm.predict_proba(scaler.transform(hog_feat.reshape(1, -1)))[0, 1]

# Run the complete CNN on the image for demo (preprocess, predict).
img_rgb = np.array(Image.open(DEMO_IMAGE).convert("RGB").resize((224, 224)))
batch = preprocess_input(np.expand_dims(img_rgb.astype(np.float32), axis=0))
cnn_prob = float(cnn.predict(batch, verbose=0)[0, 0])

if svm_prob >= 0.5:
    svm_label = "PNEUMONIA"
else:
    svm_label = "NORMAL"

if cnn_prob >= 0.5:
    cnn_label = "PNEUMONIA"
```

Held-out test image; not seen during training.

True label: NORMAL

HOG + SVM

PNEUMONIA score = 0.834

Prediction: PNEUMONIA - incorrect

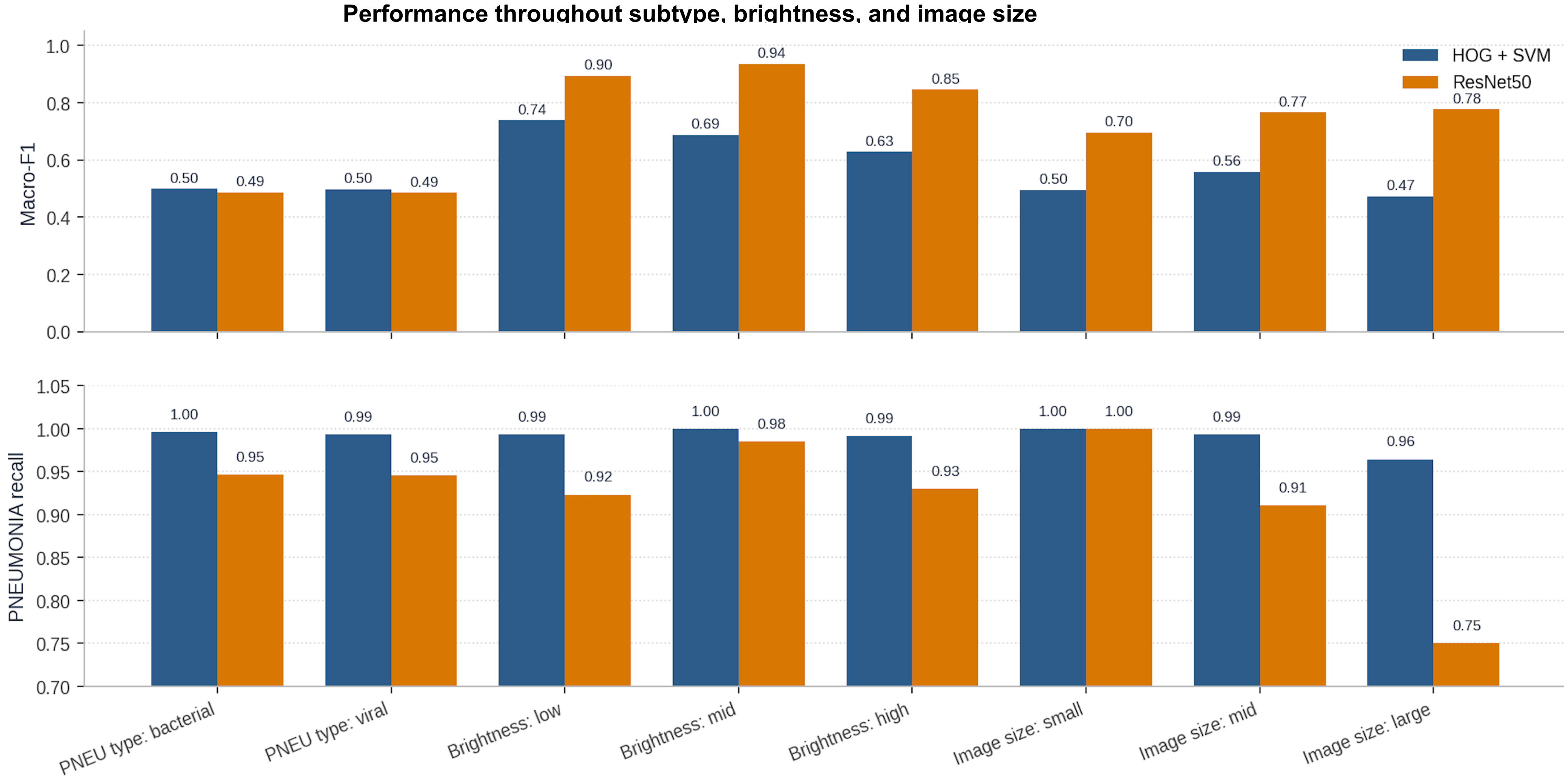
ResNet50

PNEUMONIA score = 0.051

Prediction: NORMAL - correct

One held-out example of the SVM false-alarm behaviour mentioned before.

Subgroup checks: subtype, brightness, image size



Subtype PNEU recall: SVM 0.99 ; ResNet50 0.95
Brightness macro-F1: SVM 0.63-0.74 ; ResNet50 0.85-0.94

Large image: ResNet50 PNEU recall = 0.75
Impact: Monitor subgroup performance, not only overall metrics

Sources: Buolamwini and Gebru (2018); Obermeyer et al. (2019).

Ethical considerations specific to this dataset

Privacy

- Burned-in text may remain after header removal
- Redaction and governance review required

Error costs

- False negative: missed pneumonia
- False positive: extra clinical review
- 0.5: 41 FP / 21 FN   0.8: 19 FP / 40 FN

Bias and shift

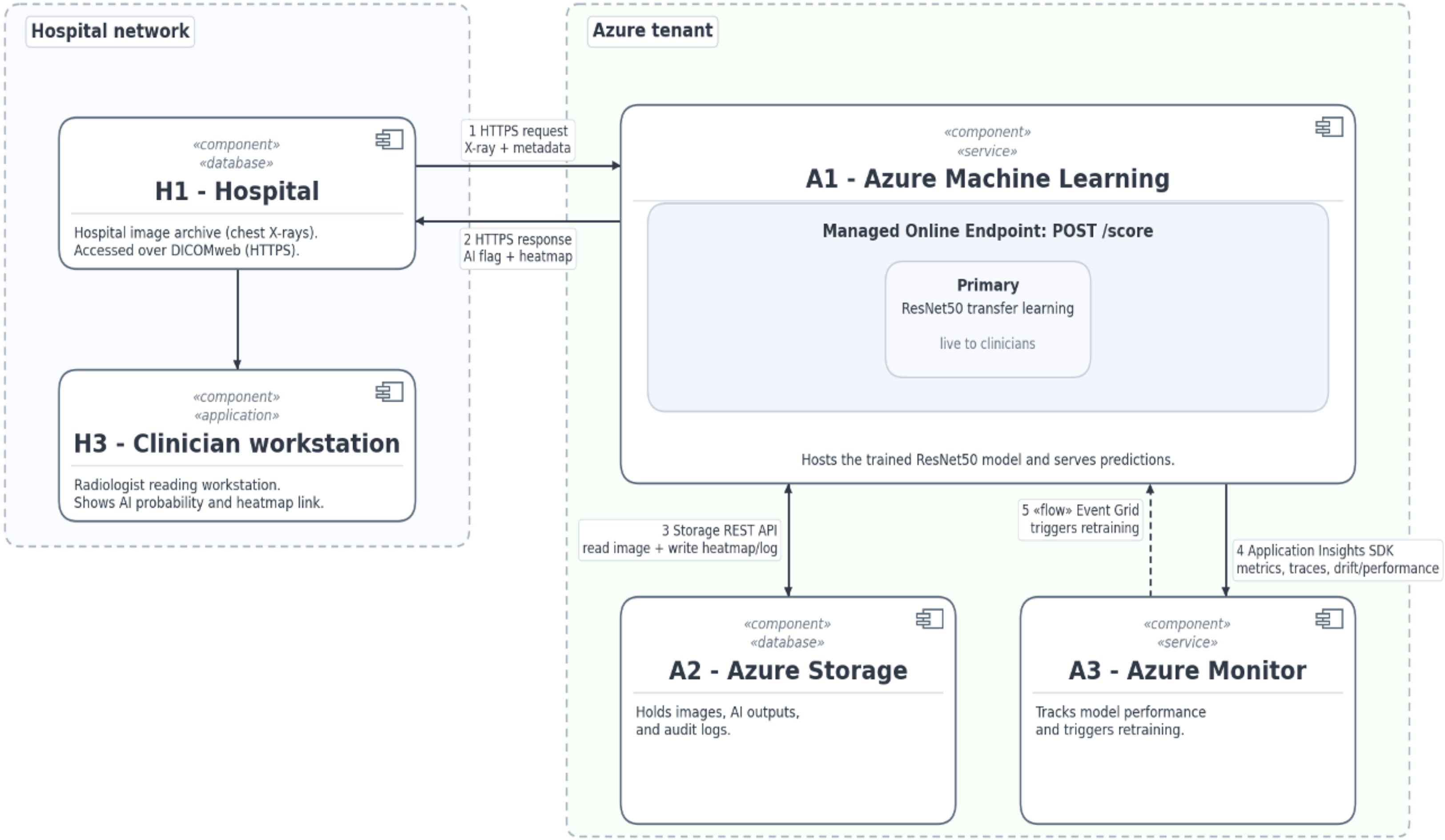
- Single-site paediatric data
- Vendor, brightness and image-size shifts
- Shortcut-feature warning from XAI

Assist to decision, no autonomous diagnosis.

Sources: DICOM Standards Committee (2011); European Parliament (2017); Hendrycks and Gimpel (2017); Leslie (2019); Provost and Fawcett (2001); Zech et al. (2018).

Possible deployment: screening assistant on Azure ML

x-ray pneumonia predictor Azure system - Component Diagram



1. PACS -> DICOMweb -> Azure ML
2. ResNet50 returns pneumonia score
3. Clinician specialist reviews in existing worklist
4. Monitor latency, drift and subgroup performance
5. Confirmed outcomes create labelled feedback
6. New model versions need validation and approval

Sources: European Parliament (2017); Microsoft (2024); Sculley et al. (2015); Sendak et al. (2020); Topol (2019).

Summary and conclusions

Why ResNet50

- Macro-F1 0.892 vs 0.683
- NORMAL recall 0.82 vs 0.36
- At 0.99 PNEU recall 102 vs 149 false alarms

Main limitations

- Large-image PNEU recall 0.75
- Shortcut warning in XAI
- Single-site paediatric dataset

Proceed Conditions

- Threshold selected on separate validation data
- Subgroup + drift monitoring
- Paediatric scope through age metadata
- Decision support

ResNet50 is the selected model as a “decision assist and inform” tool ; cannot be taken purely as an engineering or technical decision. Clinician body must remain in control.

Sources: Leslie (2019); Sculley et al. (2015); Sendak et al. (2020); Zech et al. (2018).

What I learned, next steps, and close

What I learned

- Keras fine-tuning must be verified
- CNN reproducibility is harder than classical ML
- Saved models make results traceable
- Heatmaps are hints and directional but not causal evidence
- Clinical context is paramount

Next steps

- Rerun fine-tuning correctly
- Compare DenseNet121
- Validate first on another paediatric hospital dataset and adults if possible
- Add calibration

Sources: Guo et al. (2017); TensorFlow (2024).

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